Finite-Quantum Emission Timing and Recording Costs in WRRA

Working calculation v0.5 | Wonsik Choi and Jeongin Choi | Independent researchers | 20 October 2026

Verified inputs → WRRA transformation → outputs → falsification

Verified inputs. Preserve the fixed v0.3/v0.4 reservoir beta, rate Gamma=ln2, proper-time pulse, calibrated time unit and actual 128-cell SOURCE seed. Packet resolution 256, recording cost 1% and one record token per event are new declared closure parameters. They are not measured fundamental constants. No output amplitude or matter ratio is refitted. Reproduction of known values is a valid consistency achievement; the structural contribution is the executed connection between residue, permission, finite emission and recording.

WRRA transformation. Accumulate transfer credit without double-counting reservoir energy. Emit at the first eligible threshold, split each finite debit between matter and record heat, consume a record token, and evolve expansion and pressure work between events. Event times are computed from the closure, not supplied as a list of fitted times.

Outputs. The baseline emits 128 packets. All actual SOURCE cells are evolved with their own event clocks and expanding energy ledgers. Detailed values and the finite-packet/small-seed distinction follow below.

Falsification. Reject or revise a closure/implementation failing event energy balance, positivity, permission, independent event roots, time refinement or the fixed-amplitude small-packet continuum limit. These are computational tests, not microscopic particle identification or observational likelihood tests.

Assessment and scope

An executable local cosmological toy model now computes finite emission times and recording-energy transfer. It shares the inherited background with v0.4; it does not insert finite jumps into the v0.4 finite-k metric, momentum and neutrino hierarchy. Independent local FLRW cells are the executed spatial approximation. Full finite-gradient event matching remains a subsequent task. The closure is deterministic; quantum mechanical amplitudes, a universal particle energy quantum and a microscopic recording Hamiltonian are not derived.

Executed equations and event clock

For a fixed reference cell use $8\pi G/3 = 1, a(0) = H(0) = 1, C = \rho_x a^4, J = \rho_{\text{record}} a^4, M = \rho_m a^3$ and $R = 1 - \beta + C + J$. The spectator photon and neutrino backgrounds are unchanged. No perturbed free-streaming hierarchy is integrated in this study.

$$q = \beta/N_q, \quad C_0 = \beta, \quad r_0 = J_0 = M_0 = 0,$$
$$\dot{r} = \Gamma(1+\xi)C \quad (0 < T < 1,\, Q = 1), \qquad 0 \le r < q.$$

The credit r measures integrated transfer eligibility; it is not energy in addition to C. The gate $Q = 1$ requires an unused record token. When the gate closes, credit accumulation pauses. A threshold exactly at $T = 1$ is included; subthreshold credit at the end is retained as inactive credit, never flushed. The pulse remains inherited input, not an emergent prediction.

At the first threshold $r = q$, apply the complete atomic event:

$$C^+ = C^- - q, \quad r^+ = r^- - q, \quad M^+ = M^- + (1-\epsilon)q/a_j, \quad J^+ = J^- + \epsilon q.$$

One record token is consumed. Thus the physical energy per reference cell is q/a_j, split into matter $(1-\epsilon)q/a_j$ and record heat $\epsilon q/a_j$. Neither the integer token nor the credit is counted as another energy store. The chosen heat destination is perfect photon-like radiation; no temperature, bit count, Landauer saturation or microscopic recording Hamiltonian is derived.

$$\Delta[(C+J)/a + M] = 0, \quad \dot{\ln a} = H, \quad \dot{H} = -\tfrac{1}{2}(4R/a^4 + 3M/a^3), \quad \dot{W} = HR/a.$$

H and a are continuous at each event; pressure and $\dot{H}$ can jump. The independently evolved H is checked against $H^2 = R/a^4 + M/a^3$; $R/a + M + W = 1$ is checked through the whole history. Local cell velocities are zero, so momentum conservation here is trivial; it does not test nonzero-momentum finite-gradient junctions.

Between events C is constant, giving an exact event clock:

$$T_j = \frac{1}{\Gamma(1+\xi)} \sum_{m=0}^{j-1} \frac{1}{N_q - m}.$$

There are at most N_q events, with positive spacing. Complete exhaustion occurs at finite $T_{N_q} = H_{N_q}/[\Gamma(1+\xi)]$, if the pulse and gate allow. This is a deterministic threshold closure, not a stochastic Born-rule derivation.

Numerical outputs and continuum recovery

Quantity	Result
Named automatic checks	444 passed
Initial reservoir beta	7.62452468358126e-07
Baseline N / cost fraction	256 / 0.01
Events during inherited pulse	128
First event, normalized proper time	0.00563552750347251
Last event, normalized proper time	0.997187739651105
First / last event, inherited calibrated seconds	0.00268882746563 / 0.47577902532
Maximum normalized event energy residual	1.56225e-16
Maximum relative Friedmann residual	2.30826e-12
Maximum absolute energy + work error	6.66134e-16
Independent root-clock time error	6.66134e-16
Matter M, including record cost	2.84439369238221e-07
Record heat J	3.81226234179064e-09
Actual seed: cells / count in every cell	128 / 128
Largest actual-seed time shift, normalized time	5.45337018876e-08
Actual-seed nonlinear log entropy RMS	4.79945736299e-09

The seconds inherit the previous calibrated unit 0.477120813264466 s; they are conditional outputs, not a new absolute-time measurement or a MeV chronology. The check count includes per-cell repetitions and numerical consistency checks, not 444 independent physical confirmations.

N	Matter relative error, zero cost	Matter relative error, 1% cost
32	0.013474344	0.013474344
128	0.00343298	0.00343298
512	0.000862326319	0.000862326319
2048	0.000215837537	0.000215837537
8192	5.39753953e-05	5.39753953e-05

The 1% branch yields approximately 1% less matter than the zero-cost packet branch, with the energy retained in record heat. Its event clock is unchanged because this chosen credit rule depends on C, not the cost destination. That is a branch-specific result. Odd N and a different pulse length are also checked to avoid relying on the exact half-inventory special case.

Finite packets resolve timing before tiny count differences

Seed amplitude	Counts + / -	Fixed-N entropy slope
0.01	129 / 127	0.792639438537
0.0001	128 / 128	0.194689351809
1e-06	128 / 128	0.194689350848
5e-08	128 / 128	0.194689347624

The smooth entropy tangent 0.723567388383 reproduces v0.3's 0.7235673850561154. At fixed N=256 tiny seeds leave the event count unchanged, giving a different response near 0.194689 through redshift-sensitive timing. The maximum actual-seed time shift is approximately 26.02 calibrated ns. The reported entropy is the nonlinear species contrast $\ln(M_j/M_0) - 3\ln(R_j/R_0)/4$, with the local scale-factor terms canceling. It is a zero-gradient contrast, not a new finite-k spectrum.

The small-seed and small-packet limits cannot be interchanged without a resolution condition. Roughly $q/\beta \ll |\xi|$ is needed to resolve count differences; this is not by itself a convergence guarantee. At fixed seed 0.01, N=8192 gives an entropy-slope error of 0.00213068 against the smooth finite-amplitude comparison. No claim is made that the actual 1e-8 seed has been resolved to that limit.

Rejecting controls and next integration

Omitting record heat gives a normalized per-event energy defect 0.01. Omitting the recipient redshift factor produces a maximum defect 0.723122. Capacity 7 allows exactly 7 events; zero permission permits no energy transfer. Invalid cost, rate and packet parameters are rejected.

Next: connect packet/cost values to verified microscopic inputs; implement conserving jumps and event-time perturbations in the complete v0.4 finite-k state; accelerate event bundles while retaining the seed/packet resolution condition. Complete CAMB injection, new CMB spectra, particle-specific production and MeV thermal history remain separate work.

Reference: Malik & Wands, Cosmological perturbations (2009), arXiv:0809.4944, used for conservation and multi-fluid context. The event rule is the present explicit closure, not imported quantum dynamics. Predecessor: https://zenodo.org/records/23490141.

Numerical diagnostics

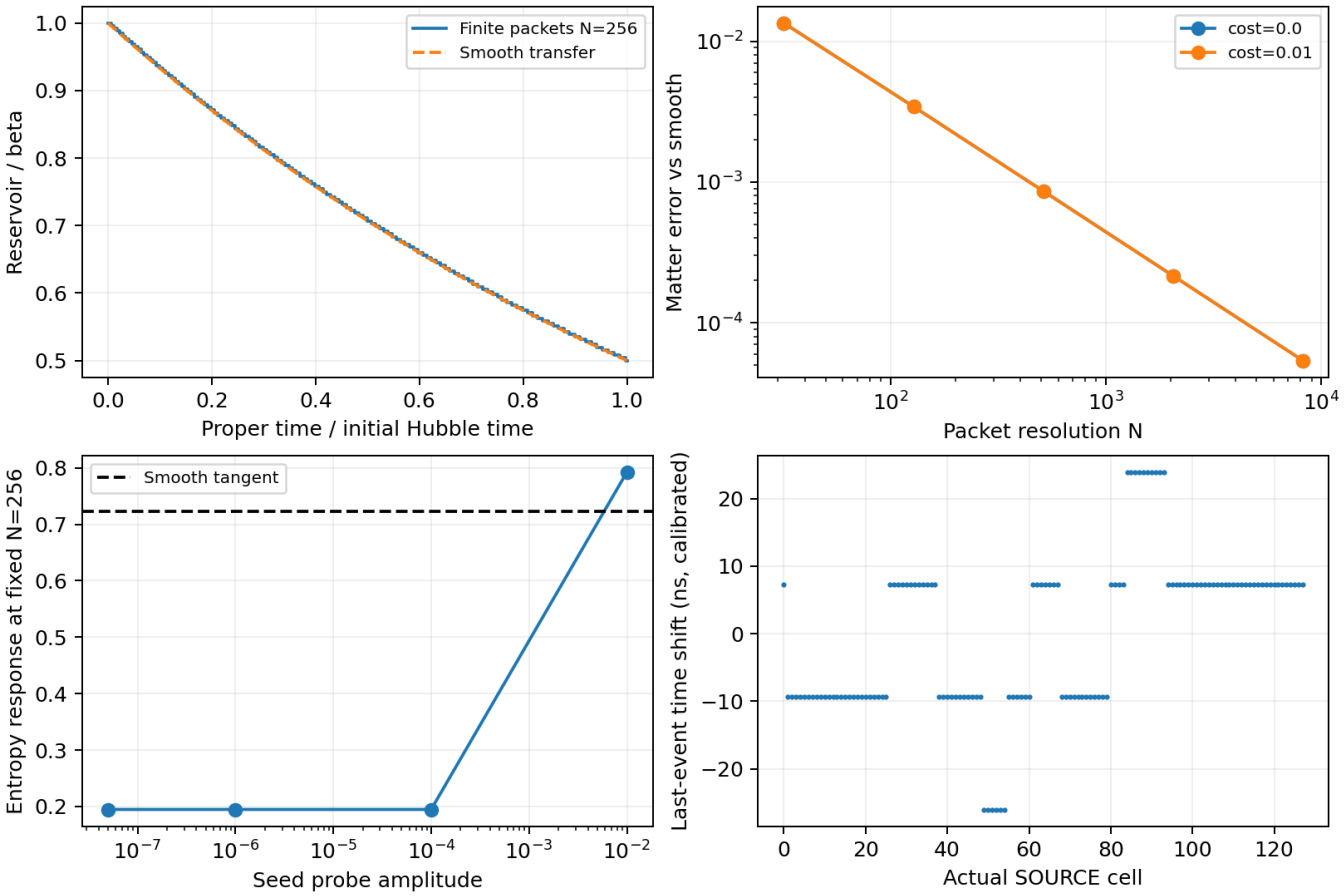


Figure 1: Staircase reservoir, smooth-limit refinement, finite-seed resolution and event-time shifts across all 128 actual SOURCE cells.

The lower-left plateau exposes the distinction between the fixed-packet small-seed response and the smooth tangent. The lower-right ns scale inherits the calibrated time unit and is a conditional timing output.